

Topic 2: HTML Basics — Structure of a Web Page

Learning Objectives

- Understand what HTML is and what it does
- Write a valid HTML document structure
- Use common tags: headings, paragraphs, links, images, lists
- Understand attributes and nesting

2.1 What is HTML?

HTML (HyperText Markup Language) is the standard language for creating web pages. It defines the **structure and content** of a page using **tags**.

HTML is NOT a programming language — it is a **markup language**. It tells the browser what to display and how content is structured, but it does not contain logic.

HTML, CSS, and JavaScript work together:

- **HTML** — Structure (the skeleton)
- **CSS** — Style (the appearance)
- **JavaScript** — Behavior (the interactivity)

2.2 HTML Document Structure

Every HTML page must have this basic structure:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First Web Page</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is my first web page.</p>
</body>
</html>
```

What each part does:

- `<!DOCTYPE html>` — Tells the browser this is an HTML5 document
- `<html>` — Root element of the page
- `<head>` — Contains metadata (not visible on page)
- `<meta charset="UTF-8">` — Character encoding for international text support
- `<title>` — Text shown in the browser tab
- `<body>` — All visible content goes here

2.3 HTML Tags and Elements

An HTML element has an opening tag, content, and a closing tag:

```
<tagname>Content goes here</tagname>
```

Some elements are **self-closing** (they have no content and no closing tag):

```
  
<br>  
<hr>  
<input type="text">
```

2.4 Headings

HTML has six levels of headings, from largest to smallest:

```
<h1>Main Page Title</h1>  
<h2>Section Heading</h2>  
<h3>Sub-section</h3>  
<h4>Sub-sub-section</h4>  
<h5>Small heading</h5>  
<h6>Smallest heading</h6>
```

Note: Use only one `<h1>` per page. It represents the main topic of the page.

2.5 Paragraphs and Text

```
<p>This is a paragraph of text.</p>  
<p>This is another paragraph. Browsers add spacing between paragraphs automatically.</p>  
  
<strong>This text is bold.</strong>  
<em>This text is italic.</em>
```

```
<br> <!-- Forces a line break within a paragraph -->
<hr> <!-- Draws a horizontal divider line -->
```

2.6 Links

```
<!-- Link to an external website -->
<a href="https://www.google.com">Visit Google</a>

<!-- Link to another page in the same site -->
<a href="about.html">About Us</a>

<!-- Opens in a new browser tab -->
<a href="https://www.google.com" target="_blank">Google (new tab)</a>
```

The `href` attribute contains the destination URL.

2.7 Images

```

```

- `src` — Path to the image file (relative or full URL)
- `alt` — Alternative text shown if image fails to load; required for accessibility
- `width` — Width in pixels (height adjusts proportionally)

2.8 Lists

Unordered list (bullet points):

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Ordered list (numbered):

```
<ol>
  <li>Install VS Code</li>
  <li>Create an HTML file</li>
```

```
<li>Open it in the browser</li>
</ol>
```

Nested list:

```
<ul>
  <li>Frontend
    <ul>
      <li>HTML</li>
      <li>CSS</li>
    </ul>
  </li>
  <li>Backend</li>
</ul>
```

2.9 HTML Attributes

Attributes provide extra information about an element. They go inside the opening tag:

```
<element attribute="value">Content</element>
```

Common attributes:

Attribute	Purpose	Example
`id`	Unique identifier (one per page)	`id="main-title"`
`class`	Reusable group name for CSS/JS	`class="btn btn-primary"`
`src`	Source file path	`src="logo.png"`
`href`	Hyperlink destination	`href="page.html"`
`alt`	Alternative text for images	`alt="Company logo"`
`style`	Inline CSS styles	`style="color: red"`

2.10 Nesting Rules

Elements must be properly nested — close the inner tag before closing the outer:

```
<!-- Correct -->
<p>Visit <a href="..."><strong>Google</strong></a> now.</p>
```

```
<!-- Wrong – overlapping tags break the page -->
<p>Visit <a href="..."><strong>Google</a></strong> now.</p>
```

Key Takeaways

- HTML defines structure and content — not appearance (that is CSS)
- Every page needs `<!DOCTYPE html>`, `<html>`, `<head>`, and `<body>`
- Tags describe the type of content (heading, paragraph, link, image, list)
- Attributes give extra information to elements
- Always close tags and nest them correctly

Reference Materials

- YouTube: Learn HTML Full Tutorial (freeCodeCamp) — <https://www.youtube.com/watch?v=kUMe1FH4CHE>
- W3Schools HTML Introduction — https://www.w3schools.com/html/html_intro.asp
- MDN HTML Basics — https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/HTML_basics

Quiz

Question 1: What does HTML stand for?

- A) Hyper Text Markup Language
- B) High Text Machine Language
- C) Hyperlink and Text Markup Language
- D) Home Tool Markup Language

Correct Answer: A — HTML stands for HyperText Markup Language. "HyperText" refers to links between pages, and "Markup" refers to the tags used to annotate content.

Question 2: Which HTML tag creates a clickable hyperlink?

- A) `<link>`
- B) `<href>`
- C) `<a>`
- D) `<url>`

Correct Answer: C — The `<a>` (anchor) tag creates hyperlinks. The destination is specified in the `href` attribute: `<a href="url">Link Text</a>`.
